

IMO(53rd) – 2012

First Day

1. Given triangle ABC the point J is the centre of the excircle opposite the vertex A. This excircle is tangent to the side BC at M, and to the lines AB and AC at K and L respectively. The lines LM and BJ meet at F, and the lines KM and CJ meet at G. Let S be the point of intersection of the lines AF and BC, and let T be the point of intersection of lines AG and BC. Prove that M is the midpoint of ST.
2. Let $n \geq 3$ be an integer, and let $a_2, a_3, \dots, a_n$ be positive real numbers such that $a_2 a_3 \dots a_n = 1$. Prove that $(1 + a_2)^2 (1 + a_3)^3 \dots (1 + a_n)^n > n^n$.
3. The liar's guessing game is a game played between two players A and B. The rules of the game depend on two positive integers k and n which are known to both players.
At the start of the game A chooses integers x and N with $1 \leq x \leq N$. Player A keeps x secret, and truthfully tells N to player B. Player B now tries to obtain information about x by asking player A questions as follows: each question consists of B specifying an arbitrary set S of positive integers (possibly one specified in some previous question), and asking A whether x belongs to S . Player B may ask as many such questions as he wishes. After each question, player A must immediately answer it with yes or no, but is allowed to lie as many times as she wants; the only restriction is that, among any $k+1$ consecutive answers, at least one answer must be truthful.
After B has asked as many questions as he wants, he must specify a set X of at most n positive integers. If x belongs to X , then B wins; otherwise, he loses. Prove that:
 - (a) If $n \geq 2^k$, then B can guarantee a win.
 - (b) For all sufficiently large k , there exists an integer $n \geq 1.99^k$ such that B cannot guarantee a win.

Second Day

4. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ (here $\mathbb{Z}$ denotes set of integers) such that, for all integers a, b, c that satisfy $a + b + c = 0$, the following equality holds:

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$
5. Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the perpendicular from C. Let X be a point in the interior of the segment CD. Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK. Show that $MK = ML$.
6. Find all positive integers n for which there exist non-negative integers $a_1, a_2, \dots, a_n$ such that

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \dots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \dots + \frac{n}{3^{a_n}} = 1.$$
